Radar-to-Candidate Pipeline TO BE 2.17.4.6.2

Status: Slices `2.17.4.6.2` and `2.17.4.7` define the implemented retrieval and evidence-backed signal-extraction boundary. Project-utility scoring and candidate assembly remain the approved target.

AS IS sources:

- `docs/architecture/RADAR_RUN_PIPELINE_AS_IS.md`;
- `docs/architecture/UPSTREAM_SEARCH_AND_SIGNAL_ARCHITECTURE.md`.

Regenerate PDF:

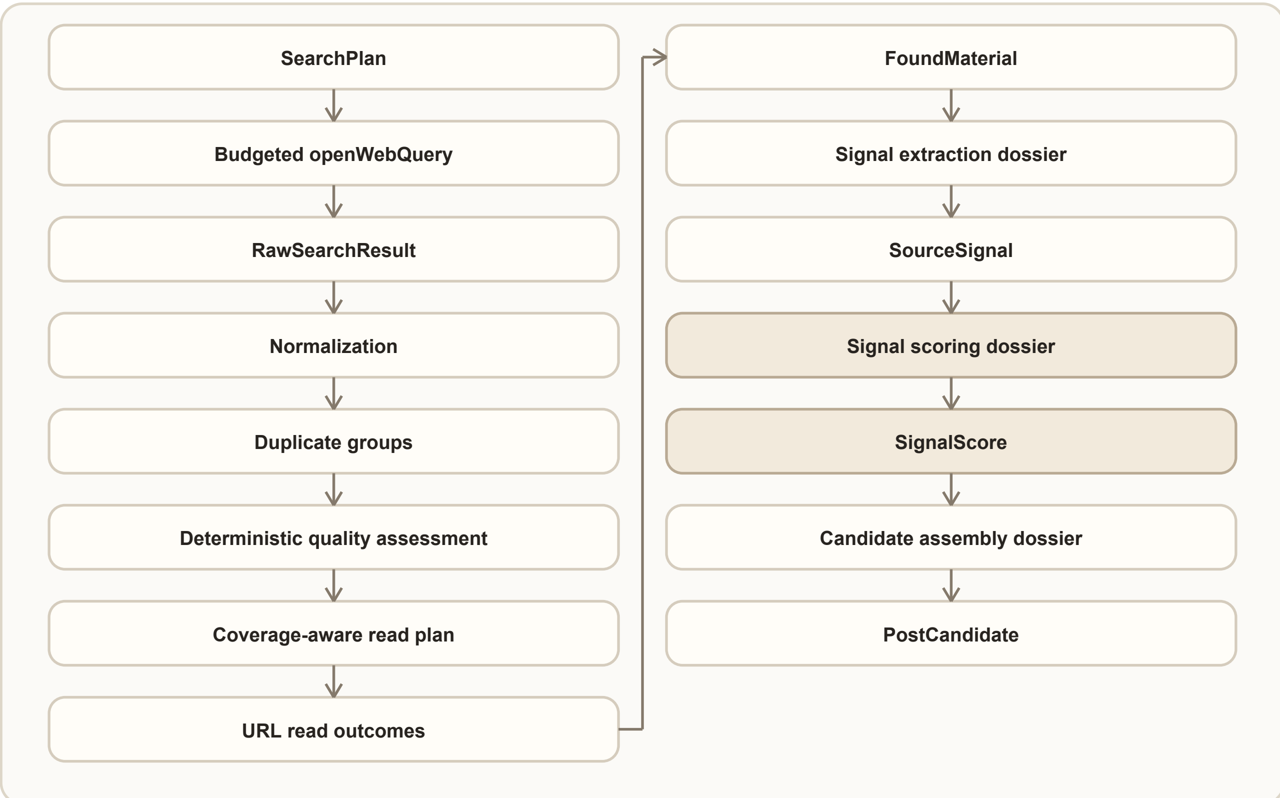
```
python scripts/generate-draft-run-pipeline-pdf.py `
  --source docs/architecture/RADAR_TO_CANDIDATE_PIPELINE_TO_BE_2_17_4_6_2.md `
  --output docs/architecture/RADAR_TO_CANDIDATE_PIPELINE_TO_BE_2_17_4_6_2.pdf
```

1. Change Intent

The current RadarRun can plan and execute web search, but its pre-read triage is too dependent on provider order, keyword overlap, exact URL equality, and a first-domain rule. The target pipeline must preserve every discovery decision while reading only the strongest bounded set of sources.

The same architecture must prevent the later signal and candidate stages from repeating the DraftRun context-growth problem. Rich upstream artifacts remain stored for replay and accountability. A provider receives only an operation-specific, budgeted projection with handles back to those artifacts.

2. Target Flow



Nodes through `FoundMaterial` are implemented by Slice `2.17.4.6.2`. Slice `2.17.4.7` implements `Signal extraction dossier -> SourceSignal`. Scoring, candidate assembly, and candidate ranking remain `NOT THIS`

`SLICE` and must be implemented by their tracker-backed slices.

3. AS IS to TO BE Mapping

Item	Status	TO BE	Proof
Search campaign planning	UNCHANGED	Deterministic intents, queries, source strategy, and budget skips remain authoritative.	Existing planner tests and trace.
Provider search	CHANGED vs AS IS	Every `openWebQuery` has a direct current-call budget and final serialized-message proof.	Operation trace, boundary tests, architecture smoke.
Citation normalization	CHANGED vs AS IS	URL, title, and snippet are bounded and normalized without deleting meaningful query parameters.	Normalization tests.
Duplicate handling	CHANGED vs AS IS	Stable duplicate groups retain all query, intent, family, and evidence handles.	Permutation and duplicate-group tests.
Pre-read scoring	CHANGED vs AS IS	Six deterministic dimensions and an explicit quality floor replace keyword-order selection.	Policy tests and readable score trace.
Read allocation	CHANGED vs AS IS	Required-family coverage is allocated first, then quality and bounded diversity.	Allocation and stress tests.
Read failure	CHANGED vs AS IS	Failed reads are failed operations; metadata-only material is not treated as readable.	Integration tests and live trace.
Read-format capability	NEW	The read plan does not spend a slot on an obvious unsupported PDF; unexpected binary responses fail safely into metadata-only evidence.	Reader capability tests and final live trace.
Found material provenance	NEW	`discoveryTrace` stores resolvable handles without copying rich snippets or full trace objects.	Handle-resolution tests.
Evidence fragments	NEW	Readable materials retain bounded, hashed fragments with offsets before full page text is discarded.	Fragment stability, bounds, and replay tests in Slice `2.17.4.7`.
Signal extraction	CHANGED vs AS IS	A backend-owned provider operation receives a bounded extraction dossier, validates exact grounding, and emits zero or more candidate SourceSignals.	Recorded benchmark, provider trace, retry replay, and live proof in Slice `2.17.4.7`.
Extraction retry	NEW	A new extraction revision reuses persisted fragments and cannot repeat search or URL reading.	API integration and idempotency proof in Slice `2.17.4.7`.
Signal scoring	NOT THIS SLICE	Future scoring receives a bounded signal dossier and direct budget proof.	Slice `2.17.4.7.1`.
Candidate assembly and ranking	NOT THIS SLICE	Future assembly receives bounded approved-signal projections.	Slices `2.17.4.8` and `2.17.4.8.1`.
Cross-run search memory	NOT THIS SLICE	Reuse of discovered results is owned by a separate durable memory policy.	Slice `2.17.4.6.6`.

4. Search Result Triage Contract

The deterministic triage chain is:

`rawResults -> normalized candidates -> duplicate groups -> dimension scores -> read plan -> read outcomes`.

Each raw result receives exactly one terminal decision:

- `selected`;
- `rejected`;
- `duplicate`;
- `invalid`;
- `deferredByBudget`.

No result may disappear because of list slicing, duplicate replacement, provider order, or an exception.

4.1 Bounded candidate projection

The triage projection contains only:

- URL, capped at 2048 characters;
- title, capped at 300 characters;
- snippet, capped at 1200 characters;
- query, intent, family, and evidence handles;
- deterministic score dimensions and short reason codes;
- diagnostic explanation, capped at 320 characters.

Full page bodies, complete search responses, source ledgers, previous operation envelopes, and nested budgets are forbidden.

4.2 Normalization and duplicates

Canonical URL normalization removes only known tracking parameters: `utm_*`, `gclid`, `fbclid`, and `ref`. Other query parameters remain because they may identify different documents or views.

Duplicate groups are stable under input permutation. They may be formed by canonical URL, tracking variants, or normalized title/snippet fingerprints. The representative is chosen by deterministic quality and lexical tie-break rules, never by provider position. A group preserves the union of all discovery handles.

4.3 Quality dimensions

Each representative receives scores from 0 to 100 for:

- relevance;
- evidence fit;
- project fit;
- source-quality signals;
- novelty;
- noise risk.

The ordering score is:

`0.30 relevance + 0.20 evidence fit + 0.20 project fit + 0.15 source quality + 0.15 novelty - noise penalty up to 30.`

The quality floor is 45. Unknown domains are neutral. Vendor sources are not rejected only for being vendors; pricing and generic promotional noise are penalized through observable content signals.

4.4 Coverage-aware reading

The read allocator first gives every executable required family the best available representative above the quality floor. Remaining capacity is filled by score. Within a score difference of 10, a new evidence type and then a new domain are preferred. Diversity never promotes a candidate below the quality floor.

Read caps remain `1/2/4` for `smoke/standard/full`. Any required direction that could not receive a read is listed in `readCoverageGaps` with a stable reason.

5. Read Outcome and Material Contract

A successful URL read creates a normal readable `FoundMaterial`. A failed read:

- produces a failed URL-read operation;
- preserves the discovery metadata in a `metadataOnly` material;
- records a structured read outcome and warning;
- does not count as a successful readable material.

If every produced material is metadata-only, RadarRun status is no better than `partial`.

`FoundMaterial.discoveryTrace` stores IDs and handles only: raw-result IDs, query IDs, intent IDs, families, evidence types, duplicate-group ID, decision reason, and read outcome. It does not copy snippets, page bodies, or the full `searchTriage` report.

5.1 Evidence fragment persistence

Before normalized page text is discarded, the reader derives bounded `contentFragments`. Each fragment has a stable ID, ordinal, text, normalized-text offsets, hash, and semantic kind. Fragments are the smallest persisted evidence unit that a signal may cite. The complete page body remains forbidden in downstream provider input.

A legacy material without fragments may expose its bounded summary as one synthetic fragment. This path is marked `legacy-summary-only`, has `DEGRADED` readiness, and cannot produce a trusted high-confidence signal. `metadataOnly`, unreadable, and empty materials are never sent to the extraction provider.

6. Provider Input Budget Boundary

`openWebQuery` is the first upstream provider-heavy operation governed by a direct budget contract.

Per call:

Measure	Limit
Query text	1000 characters
Provider input	1500 characters
Serialized messages	4000 characters
Approximate input tokens	1000

Per RadarRun:

Mode	Input characters	Approximate input tokens	Max results per query
smoke	4000	1000	3
standard	12000	3000	5
full	20000	5000	8

The effective result count cannot exceed `OPENROUTER_WEB_SEARCH_MAX_RESULTS`. Before the provider call, the direct input gate checks the current query and run totals. After message construction, a provider-neutral message guard measures the actual serialized messages. An over-limit operation is not sent and records `provider-input-over-budget` plus a structured incident.

The operation trace contains the actual `providerInput`, `payloadBudget`, `inputStats`, `payloadStats`, `messageCharCount`, model selection, and provider usage when returned. Nested metadata from an older artifact never counts as proof.

Provider-reported prompt tokens can exceed the local message estimate because the OpenRouter web-search tool adds its own retrieval context. This value is preserved as usage/cost telemetry, but it is not confused with the Glavred-controlled provider input. Production limits for provider-owned search cost and reuse belong to Slice `2.17.4.6.6`.

7. Signal Extraction Contract

Signal extraction is a backend upstream operation owned separately from project utility scoring. It answers what evidence-backed fact, change, tension, case, data point, practice, failure mode, observation, question, or recurring pattern exists in the material. It does not choose a topic, fabula, audience, value, goal, platform, or publication channel.

The rich input is the persisted set of readable `FoundMaterial` records and their fragments. `SignalExtractionContextFactory` produces a bounded radar context from scope, active rules, source intent, evidence types, and filter references. `SignalExtractionDossierFactory` then retains only:

- material IDs, source metadata, and selected bounded fragments;
- radar rule/filter references needed to understand the search scope;
- the extraction taxonomy and required output contract;
- handles back to persisted material and fragment artifacts.

Full workspace snapshots, complete pages, topics, fabulas, plans, publication history, previous envelopes, and nested budget artifacts are `neverSendToProvider`.

7.1 Terminal material decisions

Every inspected material receives exactly one decision: `signalProducing`, `insufficient`, `duplicate`, `corroborating`, `contradiction`, `noise`, or `extractionFailed`.

One material may produce zero, one, or several signals. Several materials may support or contradict a canonical signal. Signal count is never a target. Every accepted signal must resolve to at least one exact retained fragment.

7.2 Grounding and recovery

`SignalGroundingPolicy` rejects unknown handles, quotations absent from retained fragments, changed numbers or dates, unsupported certainty, and invented actors, mechanisms, outcomes, or limitations. A malformed primary response is repaired by the same model using only structured errors, then attempted with the backup model. If all provider paths fail, the terminal fallback emits no substantive signals and marks the affected materials `extractionFailed`.

Retrieval and extraction statuses are independent. Successful search and reading remain successful when extraction is partial, failed, or not run. A manual retry creates a new report revision from persisted fragments and performs no search or URL read operations.

7.3 Extraction budget boundary

Mode	Materials	Fragments per material	Fragment characters	Provider input	Serialized messages	Approximate input tokens	Max output tokens
smoke	1	3	700	6000	9000	2250	1200
standard	2	4	900	12000	16000	4000	2200
full	4	5	900	24000	30000	7500	3500

Primary, repair, and backup attempts each pass a direct current-call input gate and a final serialized-message guard. Repair context is at most 1200 characters and is included in both checks. An over-budget attempt never calls the provider. Actual OpenRouter usage is stored when supplied; missing provider usage remains explicitly unknown.

8. Future Provider Context Rule

Every future upstream provider-heavy stage must declare before implementation:

1. a typed rich input artifact;
2. an operation-specific dossier/input owner;
3. `mustHave`, `shouldHave`, `diagnosticOnly`, and `neverSendToProvider` fields;
4. a direct current-call budget profile;
5. a final serialized-message guard;
6. handles back to persisted artifacts;
7. a stress test proving bounded growth;
8. a trace-safe outcome and fallback/incident policy.

Architecture smoke rejects a new operation that does not satisfy this inventory or carry an explicit tracker-backed debt exception.

9. Trace Contract

RadarRun keeps existing fields and adds `searchTriage`:

- policy version;
- normalized candidates and dimension scores;
- duplicate groups;
- read plan and terminal decisions;
- required-family coverage and gaps;
- read outcomes;
- terminal decision counts.

Existing `rawResults`, `selectedForRead`, and `rejectedBeforeRead` remain compatible. Old runs without `searchTriage` remain readable in the UI.

Slice `2.17.4.7` additionally stores `run.signalExtraction`, `signalExtractionReport`, and `sourceSignals`. The report includes revision history, material decisions, grounding violations, duplicate/corroboration/contradiction links, provider attempts, direct budgets, final message sizes, usage, and suppressed fields. `FoundMaterial.contentFragments` are persisted evidence artifacts, not copied trace prose.

10. Success Criteria

- Every raw result has one terminal decision.
- Selection and duplicate representatives are invariant under provider result order.
- No duplicate group schedules more than one URL read.
- Required-family coverage is maximized inside the read cap and gaps are explicit.
- Known generic-news and pricing noise does not displace a suitable alternative.
- Failed reads remain failed and metadata-only results are not counted as readable.
- One hundred raw results cannot grow provider calls, provider messages, or read count beyond the active profile.
- `openWebQuery` has direct input and final-message budget proof.
- Retrieval may create candidate `SourceSignal` artifacts only through the extraction owner. It still creates no `PostCandidate`, plan slot, editorial work item, or DraftRun.
- Every inspected material has one terminal extraction decision and every accepted signal resolves to exact material and fragment handles.
- Manual extraction retry is idempotent and never repeats search or URL reading.
- Unsupported certainty, altered numbers/dates, and unresolved evidence handles are zero in the accepted benchmark and live proof.
- Recorded and comparable live proof show no quality regression relative to the pre-change industrial-AI baseline.

11. Implementation Status

- Slice `2.17.4.6.2`: triage v2, selective reading, read-outcome trace, upstream budget boundary, UI trace, recorded/live proof. `IMPLEMENTED`.
- Slice `2.17.4.7`: evidence fragments, bounded provider extraction, grounding, material decisions, retry, UI trace, recorded/live proof. `IMPLEMENTED`.
- Signal scoring, candidate assembly, candidate ranking, and cross-run search memory: `NOT THIS SLICE`.

12. Implementation Proof

The final live proof on 2026-07-13 returned 52 raw results and exactly 52 terminal decisions: 2 selected, 7 rejected, 17 duplicate, and 26 deferred by budget. It built 35 stable duplicate groups and produced two readable materials from two domains with no read warnings, accepted noise, or metadata-only output.

All three `openWebQuery` operations were directly budgeted. Serialized messages used 1,185 characters in total, local approximate input was 297 tokens, and no budget incident was recorded. The benchmark remained `warning` only because the existing three-query campaign budget skipped required `limitationCritique`; required coverage was not lost by triage.

Trace-safe pre/post evidence is committed in `docs/evidence/radar-runs/2.17.4.6.2/COMPARISON.md` and `comparison.json`.

The accepted extraction proof on 2026-07-14 is `radar-run-ai-pattern-radar-industrial-cases-8`. It returned 60 raw results, read two materials from two domains, and created three grounded signal candidates. Both materials received a terminal extraction decision; unresolved evidence handles and downstream artifacts were zero.

The initial extraction used 12,496 serialized characters against the 16,000 standard cap and 3,743 provider-reported tokens. A forced retry did not add search or URL-read operations. Its first response was rejected for an ungrounded number, same-model repair was accepted, and both attempts remained below the message cap. The live retrieval benchmark stayed `warning` only because the existing three-query budget skipped `limitationCritique`.

Trace-safe evidence is committed in `docs/evidence/radar-runs/2.17.4.7/BASELINE.md` and `LIVE_PROOF.md` with JSON peers.